Chi-Yao Huang

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Personal Website
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SUMMARY

Ph.D. researcher in representation learning, embodied AI, and world models. My research develops interpretable and controllable latent-space learning for embodied AI, connecting spatial perception, semantic understanding, and reasoning through structured visual representations. Previously a Senior R&D Engineer at HTC VIVE, where I developed core VO/SLAM tracking systems deployed across major commercial VR/AR product lines.

TECHNICAL SKILLS

Research: World models, representation learning, spatial intelligence, generative models, foundation models

Computer Vision & Robotics: SLAM, visual odometry, 3D reconstruction, sensor fusion, path planning, aerial robotics, humanoid robotics

Programming: Python, PyTorch, OpenCV, ROS, C/C++

EDUCATION

Ph.D. in Computer Science 2023 – PRESENT
Arizona State University, Tempe, AZ – Advisor: Prof. Yezhou Yang

M.S. in Robotics and Autonomous Systems 2021 – 2023
Arizona State University, Tempe, AZ

M.S. in Mechanical Engineering 2015 – 2017
National Taiwan University, Taipei, Taiwan

B.S. in Mechanical Engineering 2010 – 2014
National Sun Yat-Sen University, Kaohsiung, Taiwan

SELECTED PUBLICATIONS

Spatial World Model: Spatially Structured Latent Dynamics and Planning
Chi-Yao Huang et al. *Under review at ICLR 2027*. Developed a reward-free world model whose latent space reflects the spatial structure of the physical environment, enabling interpretable dynamics and efficient goal-directed planning.
(Web | Paper | Code | Talk)

Domain Expansion: A Latent Space Construction Framework for Multi-Task Learning
Chi-Yao Huang et al. *ICLR 2026*. Proposed an orthogonal latent-space construction framework that decomposes task-specific factors into structured subspaces to reduce representation collapse.
(Web | Paper | Code | Talk)

VOCAL: Visual Odometry via ContrAstive Learning
Chi-Yao Huang et al. *WACV 2026*. Proposed contrastive representation learning that organizes latent representations according to camera motion, producing interpretable motion-aware structure for visual odometry.
(Web | Paper | Talk | Demo)

Event-Guided Dynamic Reconstruction under Extreme Motion Blur
Arpitsinh Vaghela, **Chi-Yao Huang**, et al. *NeurIPS 2026 Workshop on Physical World AI*. Developed an event-guided reconstruction framework that jointly optimizes continuous camera trajectories and dynamic 3D Gaussians for robust reconstruction under extreme motion blur.
(Web | Paper)

ONGOING RESEARCH

Domain World Models: Layer-Structured Latent Representations for Cross-Domain Planning
Developing a reward-free world model with layer-structured latent representations that enables agents to detect domain shifts, accurately predict future states, and plan across different environments.

Temporal World Models: Dynamic Latent Representations for Time-Varying Environments
Developing a temporal world model that explicitly captures time-varying factors in latent space, enabling agents to predict environmental changes and plan in dynamic environments.

Object-Centric Learning with Contrastive Slot Attention
Developing an object-centric representation-learning framework that integrates slot attention with contrastive learning to improve object discovery, slot consistency, and compositional visual reasoning.

INDUSTRY EXPERIENCE

Senior R&D Engineer, HTC VIVE 2017 – 2021
Advanced and Creative Team, New Taipei, Taiwan

- Developed core VIO/SLAM tracking systems deployed across VIVE COSMOS, VIVE FOCUS 3, VIVE XR Elite, and VIVE FLOW.
- **VIVE COSMOS:** Architected a multi-camera tracking system for consumer VR, achieving sub-millimeter tracking precision in controlled evaluation; developed a high-performance "mothership" SLAM system. (Product | Demo)
- **VIVE FOCUS 3:** Led prototyping for a flagship MR system and optimized a tightly coupled visual-inertial SLAM pipeline on resource-constrained embedded platforms, achieving a 4× runtime speed-up. (Product)
- **VIVE XR Elite:** Engineered low-latency VIO and obstacle-avoidance systems for mixed-reality interaction; designed core data structures for real-time spatial computing. (Product)
- **VIVE FLOW:** Designed inside-out tracking for limited-FoV sensors and developed a motion-tracking prototype integrating real-time hand tracking. (Product)
- **Spatial Mapping:** Built voxel-based obstacle mapping and semantic mapping modules for VR safety and real-world interaction.

SELECTED PROJECTS

Toyota Mothership – Toyota Research Institute of North America *(Patent Pending)*: Developed a UAV pose-estimation framework using infrared imagery and IMU sensor fusion for high-altitude drone perception under challenging lighting and weather conditions. The underlying technologies are the subject of patent-pending applications. (Demo)

3D Landmark Reconstruction for Robot Localization and Mapping: Implemented a learning-based 3D reconstruction pipeline that jointly optimizes camera trajectories and object-level semantic poses for robot localization. (Demo)

3D Reconstruction and Path Planning with Signed Distance Functions: Fused dense image information with sparse feature points for efficient SLAM; implemented voxel hashing for dense mapping and integrated reconstruction with robot path planning.

Humanoid Robot Control: Developed bipedal walking patterns on non-planar surfaces by optimizing zero-moment-point trajectories and implemented localized A* planning for obstacle navigation. (Demo)

TEACHING & RESEARCH MENTORING

Graduate Research Mentoring
Mentored graduate students throughout their thesis research, providing guidance on problem formulation, literature review, experimental design, implementation, evaluation, and technical writing.

- **Zeel Bhatt** *(OpalAI)*
  Thesis on contrastive representation learning for visual odometry and robotics.
- **Khoa (Albert) Vo** *(Ph.D. student, Purdue University)*
  Thesis on multimodal reasoning and representation learning for embodied AI.
- **Shlok Mohanty** *(Graduate Student)*
  Thesis on representation learning from imbalanced robotic task data.
- **Hoang Quy Nguyen** *(Graduate Student)*
  Thesis on object-centric representation learning for structured visual understanding and downstream reasoning.

Teaching Assistant, CSE 571: Artificial Intelligence
Arizona State University
Supported course instruction through office hours, assignment grading, technical guidance, and individualized student support.

HONORS, AWARDS & GRANTS

Research Grant, Toyota Research Institute of North America 2023 – Present

Ira A. Fulton Schools of Engineering Fulton Fellows Award, Arizona State University

Student with Distinction, Arizona State University